

PHYS 427: Lecture # 5

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Lecture Span

1. Canonical Ensemble

Finishing the previous section

Pressure

$$p = T \left(\frac{\partial S}{\partial V} \right)_{U,N} \quad (1)$$

Require that $V_{tot} = V_A + V_B$ is fixed. Then, $S_{tot} = S_A + S_B$. Then maximize S by $\partial S / \partial V = 0$. Then,

$$\partial_{V_A} S_A = \partial_{V_B} S_B \implies p_A = p_B \text{ if } T_A = T_B \quad (2)$$

Then,

$$S(U, V) = Nk_B \ln V + f(\text{other constants independent of } V) \quad (3)$$

Then,

$$p = T \frac{\partial S}{\partial V} = \frac{Nk_B T}{V} \implies pV = Nk_B T \quad (4)$$

Canonical Ensemble

We have only considered closed systems, $U, N = \text{const}$. This is the microcanonical ensemble. The likeliest macrostate has the most microstates.

More common, system is not closed. But the temperature is fixed. Here, ρ has an energy ϵ and R is a reservoir. But the combined system is closed. Therefore, $U = U_R + \epsilon$ is fixed. This is called the canonical ensemble. We assume that they both have a fixed number of particles.

Properties of the canonical ensemble

Boltzmann distribution. If ρ is in a single quantum state i with energy ϵ_i . What is the probability of being in this state? Answer: boltzmann distribution.

The fundamental assumption is that

$$P(\epsilon_i) = \frac{\Omega_{R+\rho}(\epsilon_i)}{\sum_{\epsilon_i} \Omega_{R+\rho}(\epsilon_i)} \quad (5)$$

Note,

$$\Omega_{R+\rho} = \Omega_R(U_0 - \epsilon_i) \Omega_\rho(\epsilon_i) = \Omega_R(U_0 - \epsilon_i) \quad (6)$$

This is because $\Omega_\rho(\epsilon_i) = 1$ (an assumption, one state per energy). Consider

$$\frac{p(\epsilon_1)}{p(\epsilon_2)} = \frac{\Omega_{R+\rho}(U_0 - \epsilon_1)}{\Omega_{R+\rho}(U_0 - \epsilon_2)} \quad (7)$$

Then, $S = k_B \ln \Omega \implies \Omega = \exp(S/k_B)$, then we obtain

$$\frac{\exp(S_R(U_0 - \epsilon_1)/k_B)}{\exp(S_R(U_0 - \epsilon_2)/k_B)} = \exp[(S_R(U_0 - \epsilon_1) - S_R(U_0 - \epsilon_2))/k_B] \quad (8)$$

Then,

$$S_R(U_0 - \epsilon_1) = S_R(U_0) - \left. \frac{\partial S}{\partial U} \right|_{U_0} \epsilon_1 + \dots \quad (9)$$

$$\approx S_R(U_0) - \frac{\epsilon_i}{T} \quad (10)$$

For reservoir much larger than ρ , then

$$S_R(U_0 - \epsilon_1) - S_R(U_0 - \epsilon_2) \approx -\frac{\epsilon_1 - \epsilon_2}{T} \quad (11)$$

In equilibrium, $T_R = T_\rho = T$, then

$$\frac{P(\epsilon_1)}{P(\epsilon_2)} = \exp\left(-\frac{\epsilon_1 - \epsilon_2}{k_B T}\right) \quad (12)$$

We know that $\sum_i p(\epsilon_i) = 1$. Then,

$$\sum_j \frac{p(\epsilon_j)}{p(\epsilon_i)} = \sum_j \exp(-(\epsilon_j - \epsilon_i)/k_B T) = \exp(\epsilon_i/k_B T) \sum_j \exp(-\epsilon_j/k_B T) = \frac{1}{p(\epsilon_i)} \quad (13)$$

Inverting both sides yields

$$p(\epsilon_i) = \frac{\exp(-\epsilon_i/k_B T)}{\sum_j \exp(-\epsilon_j/k_B T)} \quad (14)$$

This is the boltzmann distribution. Thus, $\exp(-\epsilon_i/k_B T)$ is called the Boltzmann factor. Definition the partition function

$$Z(T) = \sum_i \exp\left(-\frac{\epsilon_i}{k_B T}\right) \quad (15)$$

Therefore,

$$p(\epsilon_i) = \frac{\exp(-\epsilon_i/k_B T)}{Z(T)} \quad (16)$$

Reminder, ϵ_i is the energy of the system. Recall that the fundamental assumption was that every microstate is equally likely. But here, the microstate of $R + \rho$ is equally likely. Assume that $\epsilon_1 > \epsilon_2$. Then, for that, $U_0 - \epsilon_1 < U_0 - \epsilon_2$ (energy of reservoir). Then the entropy of the combined system is

$$S_{R+f} = S_R(U_0 - \epsilon_1) < S_R(U_0 - \epsilon_2) \quad (17)$$

Therefore,

$$\Omega_{R+\rho} = \Omega_R(U_0 - \epsilon_1) < \Omega_R(U_0 - \epsilon_2) \quad (18)$$

Applications

Mean Values

$$U = \langle \epsilon \rangle = \sum_i \epsilon_i p(\epsilon_i) = \frac{\sum_i \epsilon_i \exp(-\epsilon_i/k_B T)}{Z(T)} \quad (19)$$

Let $\beta = 1/k_B T$. Then

$$U = -\frac{\partial}{\partial \beta} \ln(Z(T)) = -\frac{1}{Z} \partial_\beta Z(T) \quad (20)$$

Writing in terms of T ,

$$U = \frac{k_B T^2}{Z} \partial_T Z(T) \quad (21)$$

Example

Spin 0.5 in B field. What's the average energy? First, find Z . Secondly, calculate $U = \partial_\beta \ln Z(\beta)$. Here,

$$Z = e^{\mu B/k_B T} + e^{-\mu B/k_B T} \quad (22)$$

You can also calculate:

$$U = -\mu B \frac{\sinh(\mu B/k_B T)}{\cosh(\mu B/k_B T)} = -\mu B \tanh(\mu B/k_B T) \quad (23)$$

Partition function Z and its properties

Z in the canonical ensemble versus Ω in the microcanonical ensemble.

$$Z = \sum_i \exp(\epsilon_i/k_B T) \quad (24)$$

$$\Omega = \sum 1 \quad (25)$$

With ρ , write $\epsilon_1 < \epsilon_2 < \dots$. Then, $Z(T)$ contains points which get smaller. If $k_B T < \Delta\epsilon$, then only the first few terms contribute significantly. In this limit, then we can approximate $Z(T)$ with the sum of its first few terms.

If $k_B T \gg \Delta\epsilon$, then many terms contribute. Then, use an integral instead of sum.

Comment: consider Z for independent distinguishable particles. Then,

$$\epsilon_{tot} = \sum_i \epsilon_i \quad (26)$$

No interactions (independent). Then,

$$Z_{tot} = \sum_{\epsilon_1, \dots, \epsilon_n} \exp(-\beta \epsilon_{tot}) = \prod_i \sum_{\epsilon_i} e^{-\beta \epsilon_i} = Z_1 Z_2 \dots Z_N \quad (27)$$

For indistinguishable, independent particles. Then, you cannot overcount. Then,

$$Z(T) \neq Z_1^N \quad (28)$$

Write

$$Z_{tot} = Z_1^N / N! \quad (29)$$

N independent spin-1/2 particles in B field

$$U = -\frac{\partial}{\partial \beta} \ln(Z_{tot}) = -\frac{\partial}{\partial \beta} \ln(Z_1^N) = -N \partial_\beta \ln(Z_1) = -N \mu B \tanh(\beta \mu B) \quad (30)$$

$$M = -\frac{U}{B} = N \mu \tanh(\beta \mu B) \quad (31)$$

Example: Ideal gas

N particles in this box. $k_i = \pi n_i / L_i$. Here,

$$\epsilon_i = \frac{\hbar^2}{2m} \sum_i k_i^2 \quad (32)$$

Then,

$$Z_1(T) = \sum_{\epsilon_1} \exp(-\epsilon_1 / k_B T) \quad (33)$$

Plug (32) into (33). Sum can be turned into gaussian integral.

$$Z_1(T) = \sum_{n_x} \exp\left(-\frac{\hbar^2 \pi^2 n_x^2}{2m L_x^2 k_B T}\right) \sum_{n_y} \exp\left(-\frac{\hbar^2 \pi^2 n_y^2}{2m L_y^2 k_B T}\right) \dots \quad (34)$$

Let

$$\alpha_i = \frac{\hbar^2 \pi^2}{2m L_i^2 k_B T} \implies \sum e^{-\alpha_i n_i^2} \quad (35)$$

For low T, α decays fast. For high T, α decays slow. Then, you approximate it with an integral

$$\int dn_x \exp(-\alpha_x n_x^2) = \frac{1}{2} \sqrt{\frac{\pi}{\alpha_x}} \quad (36)$$

Page 6 in lecture 5 for one particle. For n particles, take $Z_1^n / N!$.